

## **Supplementary Information**

For

### **Association between ambient air pollution and perceived stress in pregnant women**

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**Table S1.** Spearman correlations among maternal air pollutants by trimesters and entire pregnancy

|                           | PM <sub>2.5</sub> | PM <sub>10</sub> | NO <sub>2</sub> | O <sub>3</sub> |
|---------------------------|-------------------|------------------|-----------------|----------------|
| 1 <sup>st</sup> trimester |                   |                  |                 |                |
| PM <sub>2.5</sub>         | 1.00              |                  |                 |                |
| PM <sub>10</sub>          | 0.91*             | 1.00             |                 |                |
| NO <sub>2</sub>           | 0.49*             | 0.54*            | 1.00            |                |
| O <sub>3</sub>            | -0.24*            | -0.24*           | -0.50*          | 1.00           |
| 2 <sup>nd</sup> trimester |                   |                  |                 |                |
| PM <sub>2.5</sub>         | 1.00              |                  |                 |                |
| PM <sub>10</sub>          | 0.91*             | 1.00             |                 |                |
| NO <sub>2</sub>           | 0.44*             | 0.49*            | 1.00            |                |
| O <sub>3</sub>            | -0.26*            | -0.27*           | -0.52*          | 1.00           |
| 3 <sup>rd</sup> trimester |                   |                  |                 |                |
| PM <sub>2.5</sub>         | 1.00              |                  |                 |                |
| PM <sub>10</sub>          | 0.90*             | 1.00             |                 |                |
| NO <sub>2</sub>           | 0.44*             | 0.47*            | 1.00            |                |
| O <sub>3</sub>            | -0.24*            | -0.23*           | -0.53*          | 1.00           |
| Whole pregnancy           |                   |                  |                 |                |
| PM <sub>2.5</sub>         | 1.00              |                  |                 |                |
| PM <sub>10</sub>          | 0.82*             | 1.00             |                 |                |
| NO <sub>2</sub>           | 0.28*             | 0.30*            | 1.00            |                |
| O <sub>3</sub>            | -0.20*            | -0.13*           | -0.46*          | 1.00           |

\*p-value &lt; 0.05.

**Table S2.** Spearman correlations of each air pollutant with itself across three trimesters and entire pregnancy

|                           | 1 <sup>st</sup> trimester | 2 <sup>nd</sup> trimester | 3 <sup>rd</sup> trimester | Whole pregnancy |
|---------------------------|---------------------------|---------------------------|---------------------------|-----------------|
| PM <sub>2.5</sub>         |                           |                           |                           |                 |
| 1 <sup>st</sup> trimester | 1.00                      |                           |                           |                 |
| 2 <sup>nd</sup> trimester | 0.30*                     | 1.00                      |                           |                 |
| 3 <sup>rd</sup> trimester | -0.13*                    | 0.32*                     | 1.00                      |                 |
| Whole pregnancy           | 0.60*                     | 0.83*                     | 0.52*                     | 1.00            |
| PM <sub>10</sub>          |                           |                           |                           |                 |
| 1 <sup>st</sup> trimester | 1.00                      |                           |                           |                 |
| 2 <sup>nd</sup> trimester | 0.08*                     | 1.00                      |                           |                 |
| 3 <sup>rd</sup> trimester | -0.49*                    | 0.12*                     | 1.00                      |                 |
| Whole pregnancy           | 0.41*                     | 0.80*                     | 0.31*                     | 1.00            |
| NO <sub>2</sub>           |                           |                           |                           |                 |
| 1 <sup>st</sup> trimester | 1.00                      |                           |                           |                 |
| 2 <sup>nd</sup> trimester | 0.56*                     | 1.00                      |                           |                 |
| 3 <sup>rd</sup> trimester | 0.31*                     | 0.57*                     | 1.00                      |                 |
| Whole pregnancy           | 0.77*                     | 0.88*                     | 0.75*                     | 1.00            |
| O <sub>3</sub>            |                           |                           |                           |                 |
| 1 <sup>st</sup> trimester | 1.00                      |                           |                           |                 |
| 2 <sup>nd</sup> trimester | 0.12*                     | 1.00                      |                           |                 |
| 3 <sup>rd</sup> trimester | -0.61*                    | 0.11*                     | 1.00                      |                 |
| Whole pregnancy           | 0.35*                     | 0.85*                     | 0.25*                     | 1.00            |

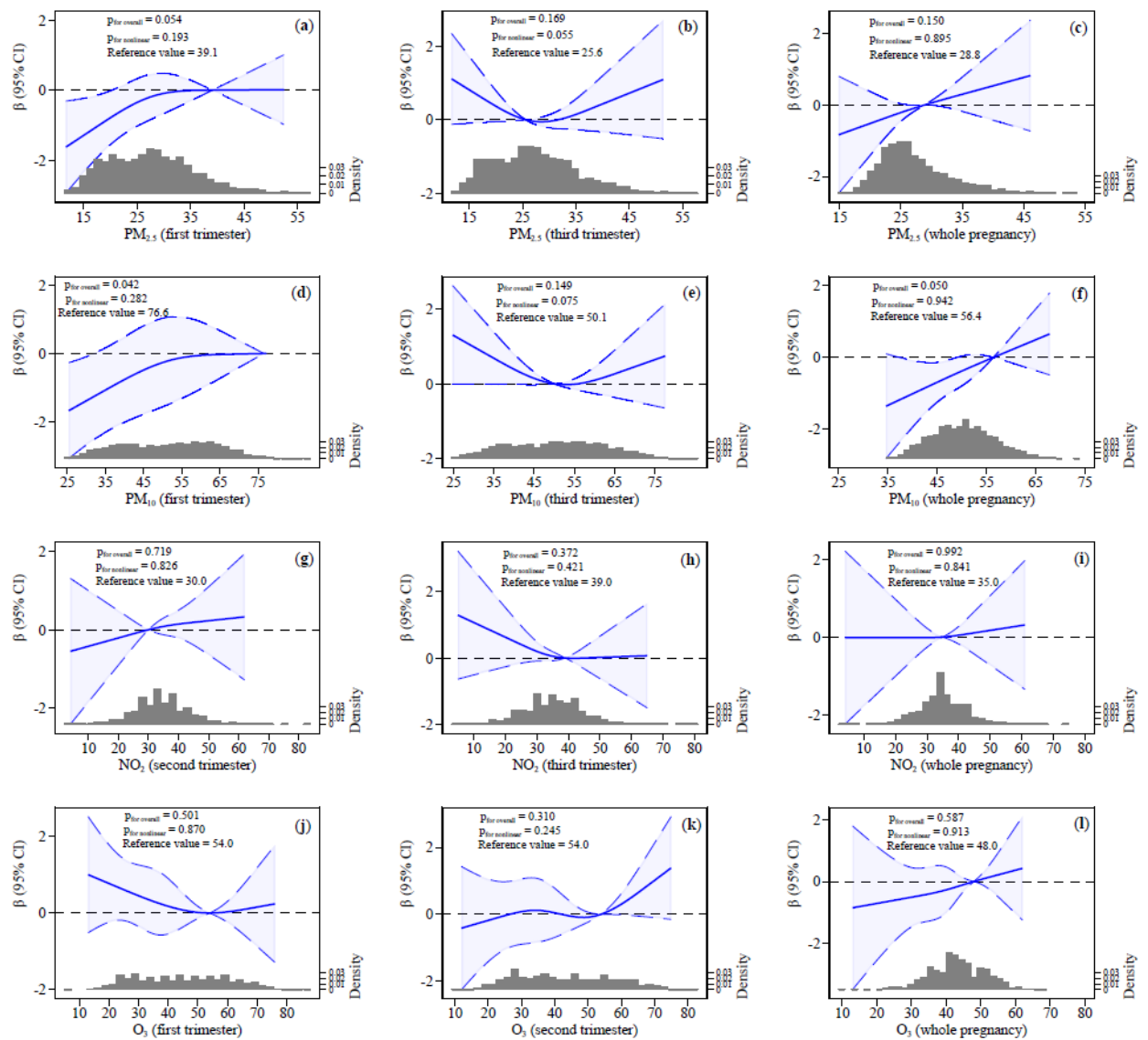
\*p-value &lt; 0.05.

**Table S3.** Associations between quartiles of maternal air pollution exposure and PSS scores

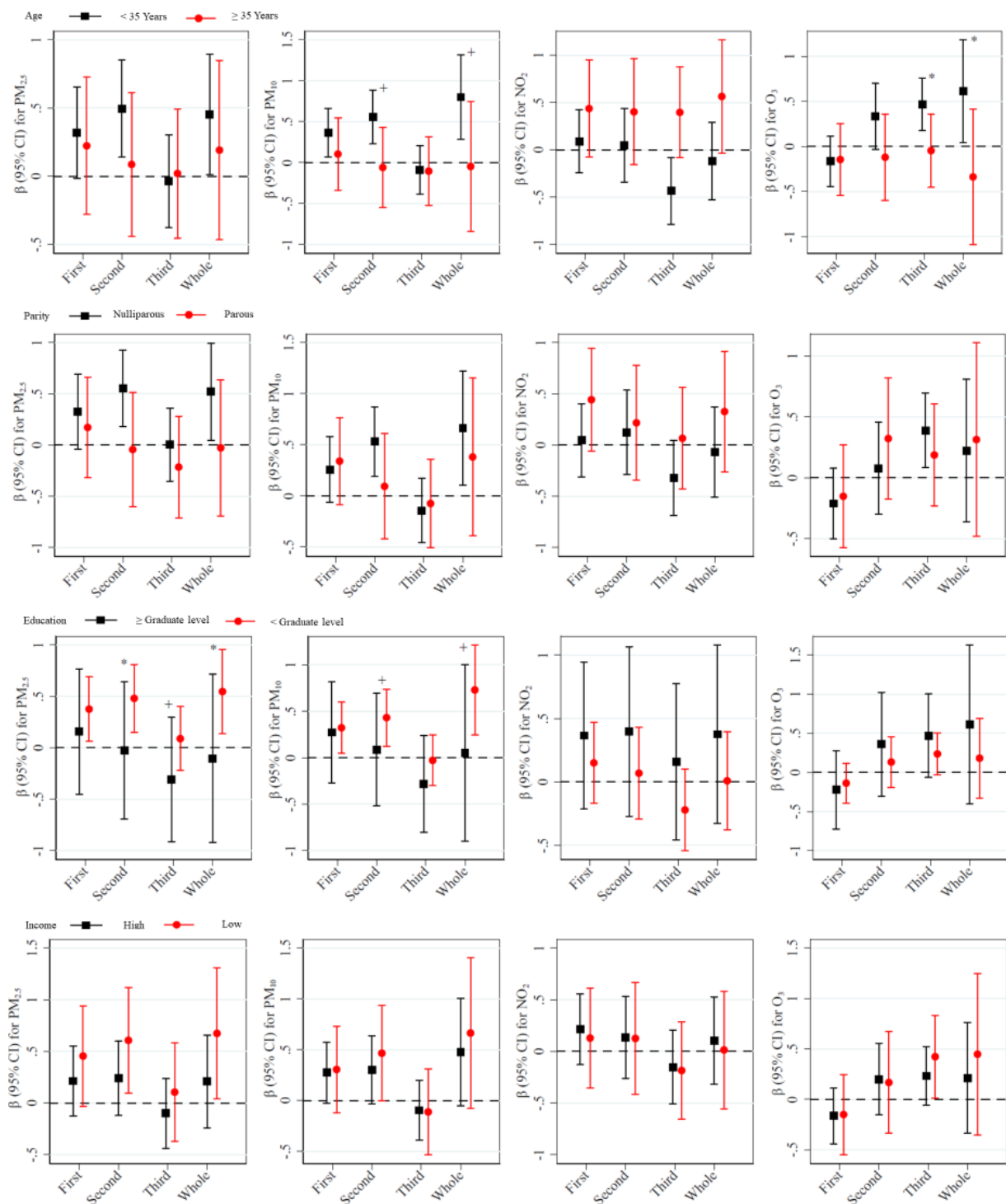
| Air pollutants           | 1 <sup>st</sup> trimester       | 2 <sup>nd</sup> trimester       | 3 <sup>rd</sup> trimester        | Whole pregnancy                |
|--------------------------|---------------------------------|---------------------------------|----------------------------------|--------------------------------|
| <b>PM<sub>2.5</sub></b>  |                                 |                                 |                                  |                                |
| 1 <sup>st</sup> quartile | Reference                       | Reference                       | Reference                        | Reference                      |
| 2 <sup>nd</sup> quartile | -0.28 (-1.14, 0.57)             | 0.09 (-0.76, 0.94)              | -0.66 (-1.50, 0.19)              | 0.73 (-0.11, 1.57)             |
| 3 <sup>rd</sup> quartile | 0.79 (-0.10, 1.68) <sup>+</sup> | 0.40 (-0.49, 1.30)              | -0.29 (-1.16, 0.58)              | 0.36 (-0.49, 1.21)             |
| 4 <sup>th</sup> quartile | 0.53 (-0.39, 1.46)              | 0.77 (-0.15, 1.69) <sup>+</sup> | -0.42 (-1.31, 0.46)              | 0.91 (0.06, 1.75) <sup>*</sup> |
| p for trend              | 0.090                           | 0.076                           | 0.104                            | 0.081                          |
| <b>PM<sub>10</sub></b>   |                                 |                                 |                                  |                                |
| 1 <sup>st</sup> quartile | Reference                       | Reference                       | Reference                        | Reference                      |
| 2 <sup>nd</sup> quartile | -0.07 (-0.93, 0.79)             | 0.47 (-0.38, 1.33)              | -0.69 (-1.54, 0.15)              | 0.22 (-0.62, 1.06)             |
| 3 <sup>rd</sup> quartile | 0.50 (-0.41, 1.41)              | 0.92 (0.03, 1.81) <sup>*</sup>  | -0.79 (-1.67, 0.09) <sup>+</sup> | 0.93 (0.08, 1.78) <sup>*</sup> |
| 4 <sup>th</sup> quartile | 0.69 (-0.26, 1.63)              | 1.07 (0.10, 2.05) <sup>*</sup>  | -0.41 (-1.33, 0.51)              | 0.72 (-0.17, 1.60)             |
| p for trend              | 0.088                           | 0.021                           | 0.358                            | 0.051                          |
| <b>NO<sub>2</sub></b>    |                                 |                                 |                                  |                                |
| 1 <sup>st</sup> quartile | Reference                       | Reference                       | Reference                        | Reference                      |
| 2 <sup>nd</sup> quartile | 0.42 (-0.39, 1.23)              | 0.04 (-0.79, 0.87)              | -0.32 (-1.16, 0.52)              | -0.22 (-1.01, 0.58)            |
| 3 <sup>rd</sup> quartile | 0.69 (-0.14, 1.53)              | 0.52 (-0.33, 1.37)              | -0.61 (-1.42, 0.20)              | 0.36 (-0.54, 1.26)             |
| 4 <sup>th</sup> quartile | 0.75 (-0.10, 1.59) <sup>+</sup> | 0.15 (-0.76, 1.06)              | -0.52 (-1.36, 0.31)              | 0.38 (-0.46, 1.23)             |
| p for trend              | 0.060                           | 0.596                           | 0.162                            | 0.234                          |
| <b>O<sub>3</sub></b>     |                                 |                                 |                                  |                                |
| 1 <sup>st</sup> quartile | Reference                       | Reference                       | Reference                        | Reference                      |
| 2 <sup>nd</sup> quartile | -0.14 (-0.98, 0.70)             | 0.13 (-0.74, 1.00)              | 0.36 (-0.48, 1.21)               | 0.86 (0.05, 1.68) <sup>*</sup> |
| 3 <sup>rd</sup> quartile | -0.58 (-1.42, 0.25)             | -0.19 (-1.19, 0.80)             | 0.92 (0.05, 1.78) <sup>*</sup>   | 0.23 (-0.64, 1.11)             |
| 4 <sup>th</sup> quartile | -0.36 (-1.25, 0.53)             | 0.71 (-0.34, 1.76)              | 1.40 (0.48, 2.32) <sup>**</sup>  | 1.00 (0.06, 1.94) <sup>*</sup> |
| p for trend              | 0.272                           | 0.209                           | 0.001                            | 0.093                          |

Quartiles of maternal air pollution exposure levels were categorized based on trimester-specific cut points in Table 2 (25<sup>th</sup>, 50<sup>th</sup>, and 75<sup>th</sup> percentiles in each trimester and entire pregnancy). p-values for trend were derived by assigning the median value of each quartile as a continuous variable. Models adjusted for maternal age, education, occupation, gestational age, maternal smoking, drinking during pregnancy, pre-pregnancy BMI, parity, season at delivery, income, asthma, thyroid disease, malignant tumors, liver disease, and hypertension or diabetes.

<sup>+</sup> p-value < 0.1. <sup>\*</sup> p-value < 0.05. <sup>\*\*</sup> p-value < 0.01.



**Figure S1.** Nonlinear effects of  $PM_{2.5}$  (a, b and c),  $PM_{10}$  (d, e and f),  $NO_2$  (g, h, and i) and  $O_3$  (j, k, and l) on the PSS score. Point estimates (solid line) and 95% confidence intervals (long dashed lines) were obtained by restricted cubic splines with three knots at the 10<sup>th</sup>, 50<sup>th</sup>, and 90<sup>th</sup> percentiles of  $PM_{2.5}$ ,  $PM_{10}$ , and  $NO_2$  distributions and four knots at the 5<sup>th</sup>, 35<sup>th</sup>, 65<sup>th</sup>, and 95<sup>th</sup> percentiles of the  $O_3$  distribution. Models adjusted for maternal age, education, occupation, gestational age, maternal smoking, drinking during pregnancy, parity, pre-pregnancy BMI, season at delivery, income, asthma, thyroid disease, malignant tumors, liver disease, and hypertension or diabetes. Histograms show the distributions of  $PM_{2.5}$  (a, b, and c),  $PM_{10}$  (d, e, and f),  $NO_2$  (g, h, and i), and  $O_3$  (j, k, and l) exposures.



**Figure S2.** Stratified analysis of air pollution and PSS scores. Adjusted difference in PSS scores per IQR increase in air pollutants stratified by age, parity, education, and income. Models adjusted for maternal age, education, occupation, gestational age, maternal smoking, drinking during pregnancy, parity, pre-pregnancy BMI, season at delivery, income, asthma, thyroid disease, malignant tumors, liver disease, and hypertension or diabetes.

+ p for interaction < 0.1. \* p for interaction < 0.05.

**Table S4.** Associations between air pollution exposure (per IQR increase) and PSS scores after excluding women with a chronic health condition

| Air pollutants                                          | Trimester | $\beta$ (95% CI)      |
|---------------------------------------------------------|-----------|-----------------------|
| PM <sub>2.5</sub> (IQR: 6.88 $\mu\text{g}/\text{m}^3$ ) | First     | 0.37 (0.07, 0.67) *   |
|                                                         | Second    | 0.43 (0.12, 0.74) **  |
|                                                         | Third     | -0.07 (-0.36, 0.23)   |
|                                                         | Pregnancy | 0.43 (0.03, 0.82) *   |
| PM <sub>10</sub> (IQR: 8.86 $\mu\text{g}/\text{m}^3$ )  | First     | 0.35 (0.09, 0.61) **  |
|                                                         | Second    | 0.42 (0.13, 0.71) **  |
|                                                         | Third     | -0.17 (-0.43, 0.10)   |
|                                                         | Pregnancy | 0.60 (0.14, 1.07) *   |
| NO <sub>2</sub> (IQR: 9.0 ppb)                          | First     | 0.16 (-0.14, 0.47)    |
|                                                         | Second    | 0.07 (-0.27, 0.42)    |
|                                                         | Third     | -0.27 (-0.57, 0.04) + |
|                                                         | Pregnancy | -0.003 (-0.37, 0.36)  |
| O <sub>3</sub> (IQR: 11.0 ppb)                          | First     | -0.17 (-0.42, 0.07)   |
|                                                         | Second    | 0.29 (-0.02, 0.61) +  |
|                                                         | Third     | 0.38 (0.12, 0.63) **  |
|                                                         | Pregnancy | 0.46 (-0.04, 0.96) +  |

n = 1854. IQR, interquartile range; CI, confidence interval. Models adjusted for maternal age, education, occupation, family income, gestational age, maternal smoking, drinking during pregnancy, pre-pregnancy BMI, parity, and season at delivery.

+ p-value < 0.1. \* p-value < 0.05. \*\* p-value < 0.01.

**Table S5.** Comparison of imputation methods using adjusted models

| Air pollutants                                          | Trimester | Simple mean substitution | Multiple imputation <sup>a</sup> |
|---------------------------------------------------------|-----------|--------------------------|----------------------------------|
|                                                         |           | $\beta$ (95% CI)         | $\beta$ (95% CI)                 |
| PM <sub>2.5</sub> (IQR: 6.88 $\mu\text{g}/\text{m}^3$ ) | First     | 0.29 (0.02, 0.56) *      | 0.30 (0.02, 0.57) *              |
|                                                         | Second    | 0.36 (0.07, 0.65) *      | 0.37 (0.08, 0.65) *              |
|                                                         | Third     | 0.01 (-0.26, 0.28)       | 0.01 (-0.26, 0.28)               |
|                                                         | Pregnancy | 0.39 (0.03, 0.74) *      | 0.39 (0.03, 0.75) *              |
| PM <sub>10</sub> (IQR: 8.86 $\mu\text{g}/\text{m}^3$ )  | First     | 0.26 (0.02, 0.50) *      | 0.27 (0.03, 0.51) *              |
|                                                         | Second    | 0.34 (0.08, 0.61) *      | 0.35 (0.08, 0.61) *              |
|                                                         | Third     | -0.08 (-0.32, 0.16)      | -0.08 (-0.31, 0.16)              |
|                                                         | Pregnancy | 0.53 (0.11, 0.95) *      | 0.54 (0.11, 0.96) *              |
| NO <sub>2</sub> (IQR: 9.0 ppb)                          | First     | 0.14 (-0.13, 0.41)       | 0.14 (-0.13, 0.41)               |
|                                                         | Second    | 0.11 (-0.20, 0.43)       | 0.11 (-0.20, 0.42)               |
|                                                         | Third     | -0.17 (0.45, 0.11)       | -0.17 (-0.45, 0.11)              |
|                                                         | Pregnancy | 0.04 (-0.29, 0.38)       | 0.04 (-0.29, 0.38)               |
| O <sub>3</sub> (IQR: 11.0 ppb)                          | First     | -0.14 (-0.36, 0.09)      | -0.14 (-0.36, 0.08)              |
|                                                         | Second    | 1.13 (-0.15, 0.42)       | 0.14 (-0.15, 0.42)               |
|                                                         | Third     | 0.25 (0.02, 0.48) *      | 0.25 (0.02, 0.48) *              |
|                                                         | Pregnancy | 0.22 (-0.22, 0.67)       | 0.22 (-0.22, 0.67)               |

n = 2227. IQR, interquartile range; CI, confidence interval. The model 4 shown in Table 3 was used for the analysis.

<sup>a</sup> Missing data was filled in using multiple imputation technique following a Markov Chain Monte Carlo distribution.

\*p-value < 0.05.

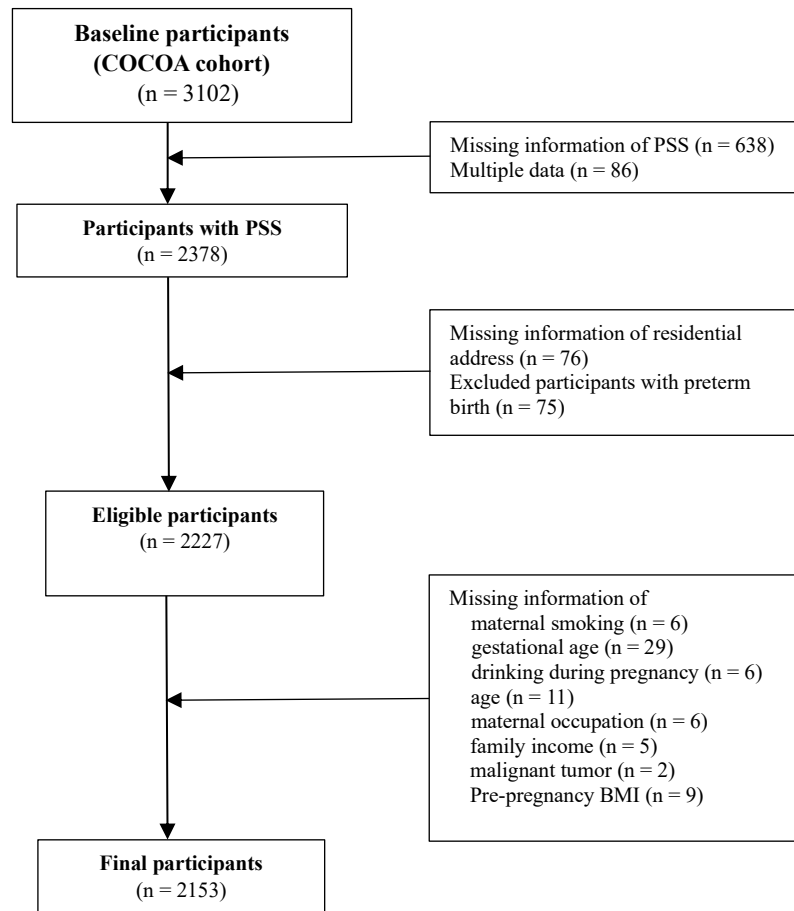
**Table S6.** Sensitivity analysis using E-value estimates

| Air pollutant                                     | Trimester | RR (95% CI) <sup>a</sup> | E-value <sup>b</sup> |
|---------------------------------------------------|-----------|--------------------------|----------------------|
| PM <sub>2.5</sub> (IQR = 6.88 µg/m <sup>3</sup> ) | First     | 1.014 (1.005, 1.023) **  | 1.13 (1.08)          |
|                                                   | Second    | 1.018 (1.008, 1.027) *** | 1.15 (1.10)          |
|                                                   | Third     | 0.998 (0.990, 1.007)     | 1.05 (1.00)          |
|                                                   | Pregnancy | 1.018 (1.006, 1.030) **  | 1.15 (1.08)          |
| PM <sub>10</sub> (IQR = 8.86 µg/m <sup>3</sup> )  | First     | 1.014 (1.006, 1.022) *** | 1.13 (1.08)          |
|                                                   | Second    | 1.018 (1.009, 1.027) *** | 1.15 (1.10)          |
|                                                   | Third     | 0.995 (0.987, 1.003)     | 1.08 (1.00)          |
|                                                   | Pregnancy | 1.027 (1.013, 1.041) *** | 1.19 (1.13)          |
| NO <sub>2</sub> (IQR = 9.0 ppb)                   | First     | 1.009 (0.999, 1.018) +   | 1.10 (1.00)          |
|                                                   | Second    | 1.006 (0.996, 1.017)     | 1.08 (1.00)          |
|                                                   | Third     | 0.992 (0.983, 1.001)     | 1.10 (1.00)          |
|                                                   | Pregnancy | 1.003 (0.992, 1.014)     | 1.06 (1.00)          |
| O <sub>3</sub> (IQR = 11.0 ppb)                   | First     | 0.993 (0.986, 1.0001) +  | 1.09 (1.00)          |
|                                                   | Second    | 1.009 (1.00004, 1.019) * | 1.10 (1.01)          |
|                                                   | Third     | 1.015 (1.007, 1.023) *** | 1.14 (1.09)          |
|                                                   | Pregnancy | 1.016 (1.001, 1.031) *   | 1.14 (1.03)          |

n = 2153. IQR, interquartile range; CI, confidence interval.

<sup>a</sup> The relative risk (RR) and 95% CIs are obtained from Poisson regression model using the model 4 shown in Table 3, indicating RR of perceived stress for interquartile range change in air pollution exposure during pregnancy.

<sup>b</sup> E-values for point estimates (RR) and corresponding CIs. Lower 95% CIs are reported in parentheses.



**Figure S3.** Final analytic sample and sample size. The figure shows how the final analytic sample used in this study was derived from the Cohort for Childhood Origin of Asthma and allergic diseases (COCOA). It shows the application of exclusion criteria based on exposure and outcome variables of interest and availability of critical variables for modeling. BMI, body mass index; PSS, perceived stress scale.

**Table S7.** Characteristics of the study participants from the COCOA cohort included and not included in the present study

| <b>Characteristics</b>                      | <b>Included (n = 2153)</b> | <b>Excluded (n = 74)</b> | <b>p-value</b> |
|---------------------------------------------|----------------------------|--------------------------|----------------|
| <b>Age (years)</b>                          | 33.1 ± 3.6                 | 32.8 ± 3.5               | 0.510          |
| <b>Pre-pregnancy BMI (kg/m<sup>2</sup>)</b> |                            |                          |                |
| < 25                                        | 2012 (93.5)                | 61 (82.4)                | 0.118          |
| ≥ 25                                        | 141 (6.5)                  | 1 (1.4)                  |                |
| <b>Parity</b>                               |                            |                          |                |
| Nulliparous                                 | 1314 (61.0)                | 38 (51.4)                | 0.555          |
| Parous                                      | 661 (30.7)                 | 16 (21.6)                |                |
| <b>History of smoking</b>                   |                            |                          |                |
| Never                                       | 1988 (92.3)                | 61 (82.4)                | 0.424          |
| Ever                                        | 165 (7.7)                  | 7 (9.5)                  |                |
| <b>Drinking during pregnancy</b>            |                            |                          |                |
| No                                          | 1989 (92.4)                | 61 (82.4)                | 0.685          |
| Yes                                         | 164 (7.6)                  | 6 (8.1)                  |                |
| <b>Occupation</b>                           |                            |                          |                |
| No                                          | 724 (33.6)                 | 28 (37.8)                | 0.137          |
| Yes                                         | 1429 (66.4)                | 38 (51.4)                |                |
| <b>Education</b>                            |                            |                          |                |
| Secondary school                            | 99 (4.6)                   | 4 (5.4)                  | 0.788          |
| College or university                       | 1579 (73.3)                | 56 (75.7)                |                |
| Graduate school                             | 475 (22.1)                 | 14 (18.9)                |                |
| <b>Gestational age (weeks)</b>              | 39.3 ± 1.1                 | 39.2 (1.0)               | 0.514          |
| <b>Family income</b>                        |                            |                          |                |
| High (≥ 4 million per month)                | 1353 (62.8)                | 34 (45.9)                | 0.031          |
| Low (< 4 million per month)                 | 800 (37.2)                 | 34 (45.9)                |                |
| <b>Asthma</b>                               |                            |                          |                |
| No                                          | 2081 (96.7)                | 73 (98.6)                | 0.344          |
| Yes                                         | 72 (3.3)                   | 1 (1.4)                  |                |
| <b>Thyroid disease</b>                      |                            |                          |                |
| No                                          | 2011 (93.4)                | 70 (94.6)                | 0.684          |
| Yes                                         | 142 (6.6)                  | 4 (5.4)                  |                |
| <b>Malignant tumor</b>                      |                            |                          |                |
| No                                          | 2123 (98.6)                | 71 (95.9)                | 0.997          |
| Yes                                         | 30 (1.4)                   | 1 (1.4)                  |                |
| <b>Liver disease</b>                        |                            |                          |                |
| No                                          | 2084 (96.8)                | 72 (97.3)                | 0.809          |
| Yes                                         | 69 (3.2)                   | 2 (2.7)                  |                |
| <b>Hypertension or Diabetes mellitus</b>    |                            |                          |                |
| No                                          | 2121 (98.5)                | 73 (98.6)                | 0.925          |
| Yes                                         | 32 (1.5)                   | 1 (1.4)                  |                |
| <b>Season at delivery</b>                   |                            |                          |                |
| Spring                                      | 465 (21.6)                 | 7 (9.5)                  | 0.089          |
| Summer                                      | 432 (20.1)                 | 18 (24.3)                |                |
| Autumn                                      | 489 (22.7)                 | 15 (20.3)                |                |
| Winter                                      | 590 (27.4)                 | 13 (17.6)                |                |

\*p values obtained from chi square test for categorical variables and t-test for continuous variables. Data are shown as n (%) or mean ± standard deviation. Of the 225 participants with PSS who were excluded, we considered 74 after removing 151 participants with missing information of residential addresses and preterm birth. Total numbers may not be equal to the total included and excluded numbers for some characteristics due to missing data. BMI, body mass index.